Transforming Global Wine Trading: A Digital Platform for South Africa

The South African wine industry faces a challenge: inefficient distribution. This platform connects producers directly with restaurants, ensuring fresh, high-quality wines reach tables throughout the world. This platform aims to streamline wine trading by connecting producers with restaurants in South Africa. It leverages data analysis and advanced technology to create an efficient marketplace for both sides.

Challenges to be addressed

1

Long Lead Times

Traditional wine distribution channels often involve lengthy delays, impacting freshness and quality.

2

Limited Reach

Producers struggle to access a wider global market, limiting their potential.

3

Price Volatility

Uncertain demand and inefficient logistics lead to price fluctuations, impacting both producers and restaurants.

4

Logistics Planning

Optimizing delivery routes based on distance, traffic, and weather conditions.

Objectives for Implementing a Wine Auction System

1

Comprehensive Data Collection

The platform will leverage web scraping and API integrations to gather comprehensive data on restaurants, flights, road conditions, and weather.

2

Advanced Trading & Auction System

The system will feature advanced trading and auction capabilities, including demand-offer matching, multi-unit auctions, and fair allocation.

3

Optimized Routing & Delivery

The platform will optimize routes and deliveries based on traffic conditions, costs, and flight schedules to ensure timely and cost-effective wine delivery.

Technological Requirements

Web Scraping & API Integration

The platform will leverage web scraping tools like BeautifulSoup, Scrapy, and Selenium to collect data on restaurants and market trends.

- Amadeus API for flight schedules and pricing
- TomTom API for real-time traffic conditions
- OpenWeatherMap API for weather forecasts

Big Data Tools

Hadoop and Spark will be used to process large volumes of data, allowing for real-time and batch processing.

- Real-time analysis of restaurant data
- Batch processing of historical data for trend analysis

Hadoop and PySpark for Advanced Analytics

Data Ingestion

The platform will leverage Hadoop's distributed storage and processing capabilities to efficiently ingest large volumes of data from diverse sources.

This includes web scraping, API integrations, and data feeds from various sources such as restaurant databases, flight booking systems, and weather APIs.

Data Visualization

Interactive dashboards and reports will be created using visualization tools like Tableau and Power BI to provide actionable insights to stakeholders.

These visualizations will help analyze market trends, track auction performance, and identify opportunities for optimizing wine distribution and logistics.

1

2

3

Data Processing

PySpark, a Python API for Apache Spark, will be used for performing complex data transformations and analysis on the Hadoop cluster.

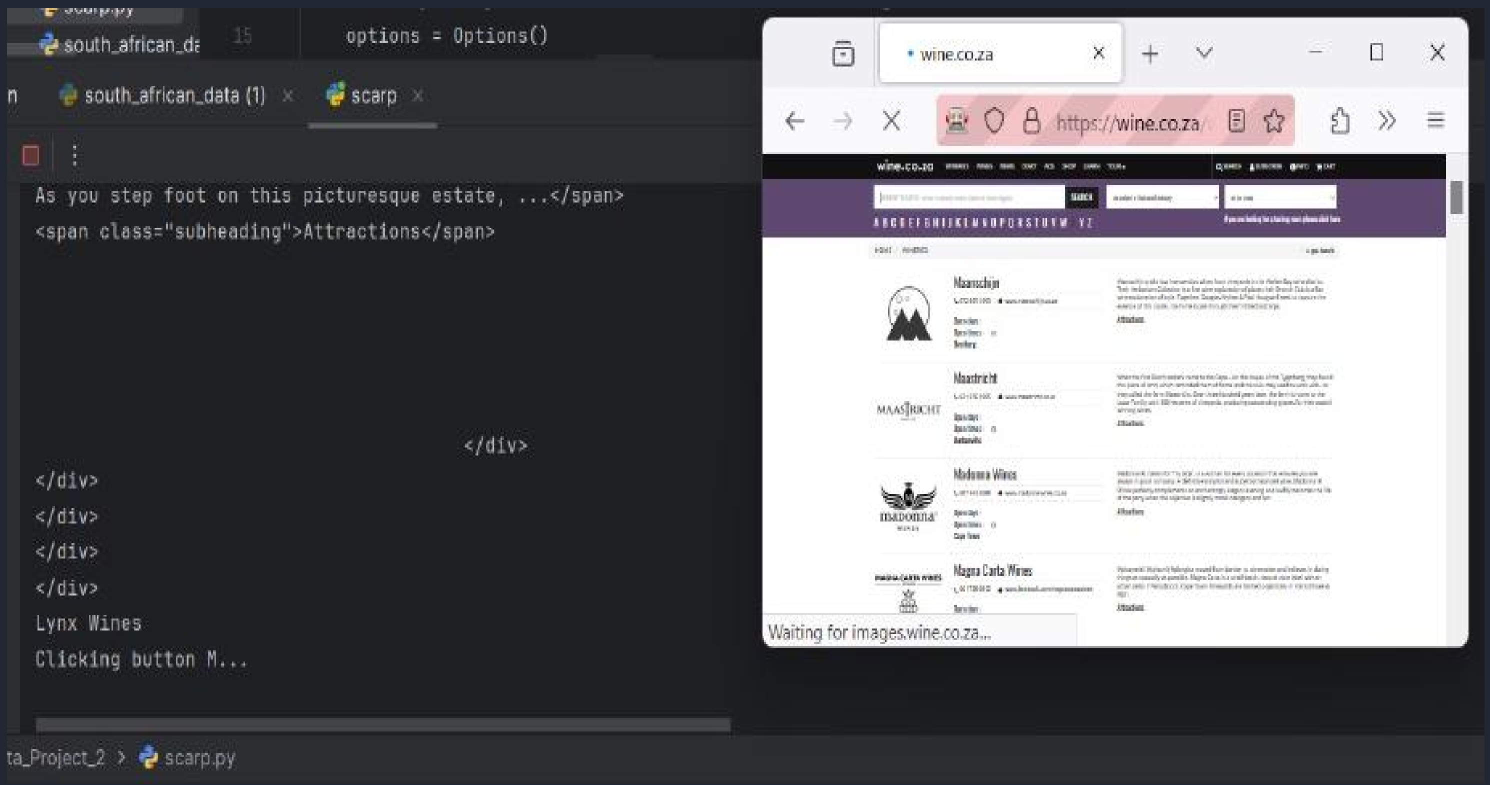
This involves cleaning, transforming, and aggregating data to derive meaningful insights and support auction algorithms.

Data Scraping & Injestion

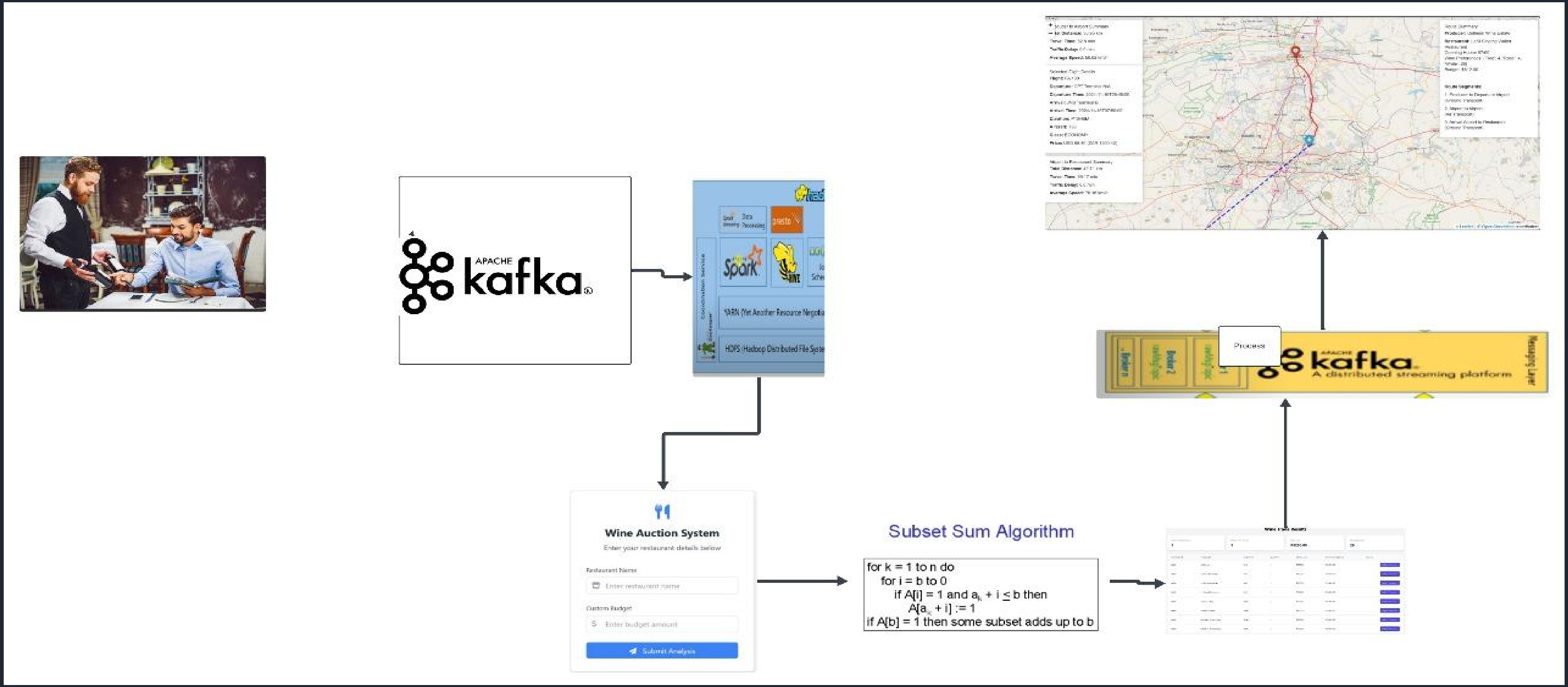
Restaurants & Wine Producers

Conducted data scraping on dining restaurants and wine producers in South Africa, extracting the following columns: Restaurant/Producer Name, Location, Contact Details, Longitude, Latitude, Nearest Airport Code, and Restaurant Wine Preferences from the following sources:

- Wine.co.za
- Restaurants.co.za

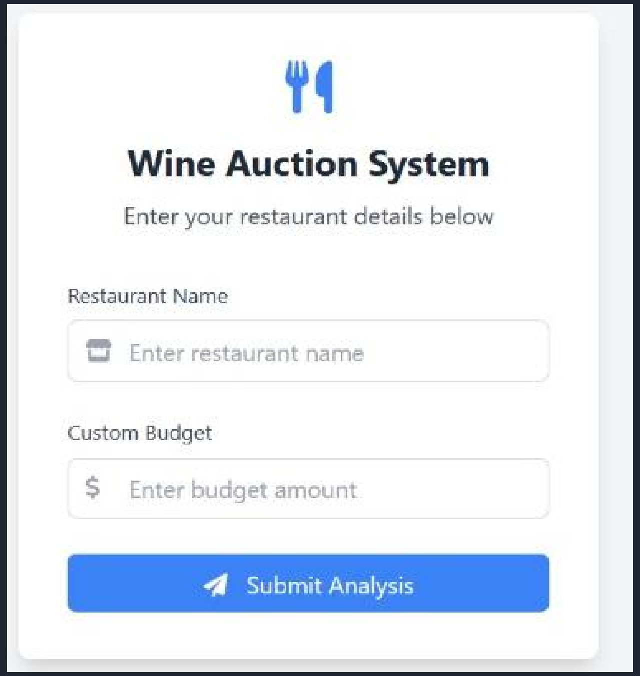


Wine Trading and Auction System Architecture



Wine Trading and Auction System

- Utilizes a multi-unit auction model (perfect-sum algorithm) to match requests with offers based on parameters (budget, wine type, quantity).
- Initiates the auction process with restaurant requests and streams auction results and delivery details to logistics.
- Real-time visualizations for insights into auction results, financial performance (cost), and delivery logistics.



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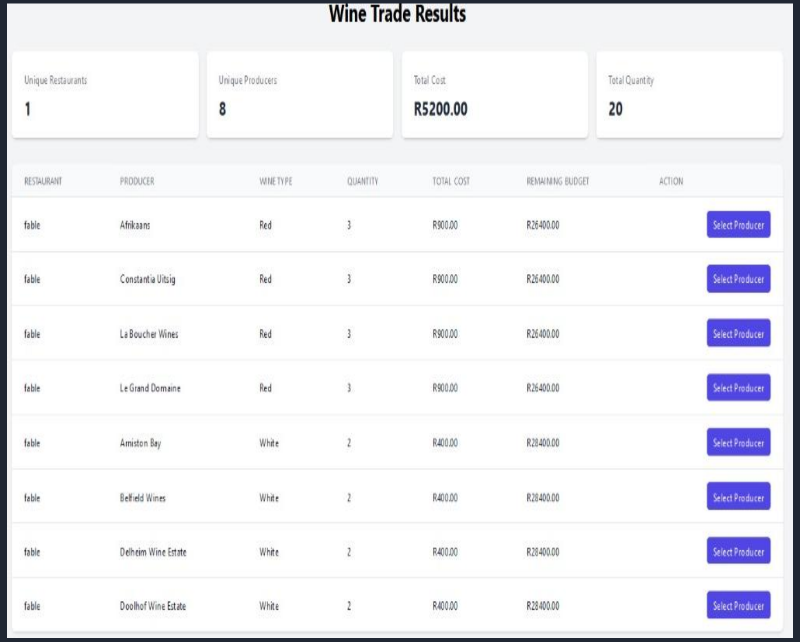

Wine Trade Results

Unique Restaurants	Unique Producers	Total Cost	Total Quantity
1	8	R5200.00	20

RESTAURANT	PRODUCER	WINE TYPE	QUANTITY	TOTAL COST	REMAINING BUDGET	ACTION
fable	Afrikaans	Red	3	R900.00	R26400.00	Select Producer
fable	Constantia Uitsig	Red	3	R900.00	R26400.00	Select Producer
fable	La Boucher Wines	Red	3	R900.00	R26400.00	Select Producer
fable	Le Grand Domaine	Red	3	R900.00	R26400.00	Select Producer
fable	Arniston Bay	White	2	R400.00	R28400.00	Select Producer
fable	Belfield Wines	White	2	R400.00	R28400.00	Select Producer
fable	Delheim Wine Estate	White	2	R400.00	R28400.00	Select Producer
fable	Doolhof Wine Estate	White	2	R400.00	R28400.00	Select Producer

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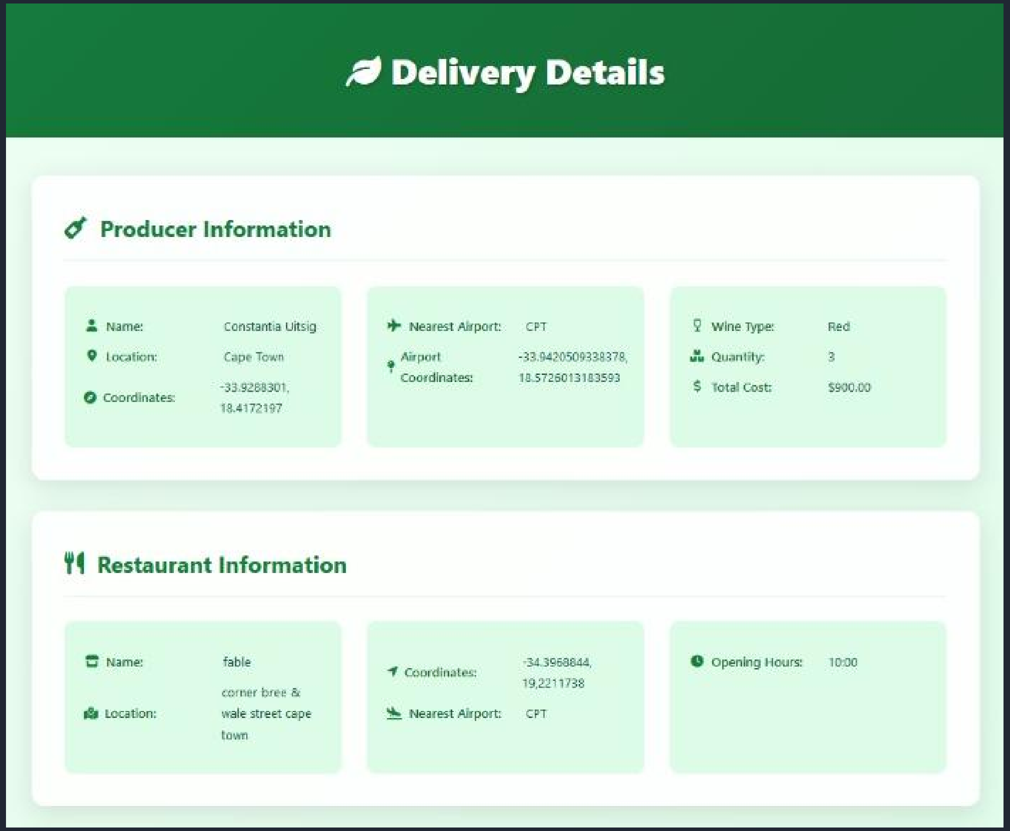

Optimized Delivery and Logistics

1 **Route Optimization**

Algorithms optimize delivery routes based on factors such as traffic conditions, weather, and restaurant locations.

2 **Fleet Management**

Real-time tracking of delivery vehicles, optimizing routes and ensuring timely deliveries.

3 **Inventory Management**

Real-time inventory tracking and forecasting to ensure sufficient supply of wines at all times.

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Total Price: USD 68.40 (ZAR 1200.42)

Itinerary 1:
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Flight: FA 200
Departure: JNB Terminal B
Departure Time: 2024-11-16 05:55 AM
Arrival: CPT Terminal N/A
Arrival Time: 2024-11-16 08:15 AM
Flight Duration: 2h 20m
Aircraft: 738
Stops: 0
Cabin Class: ECONOMY

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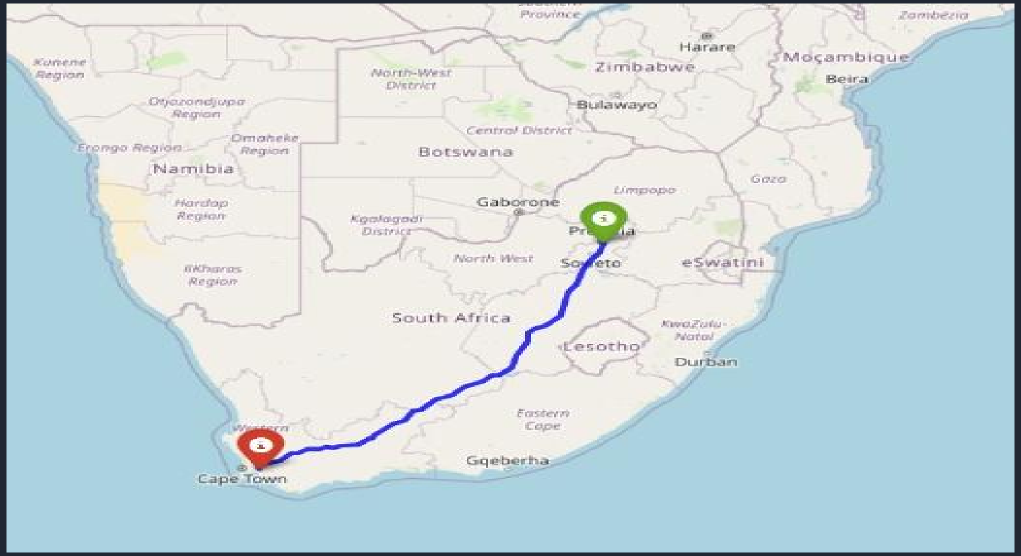
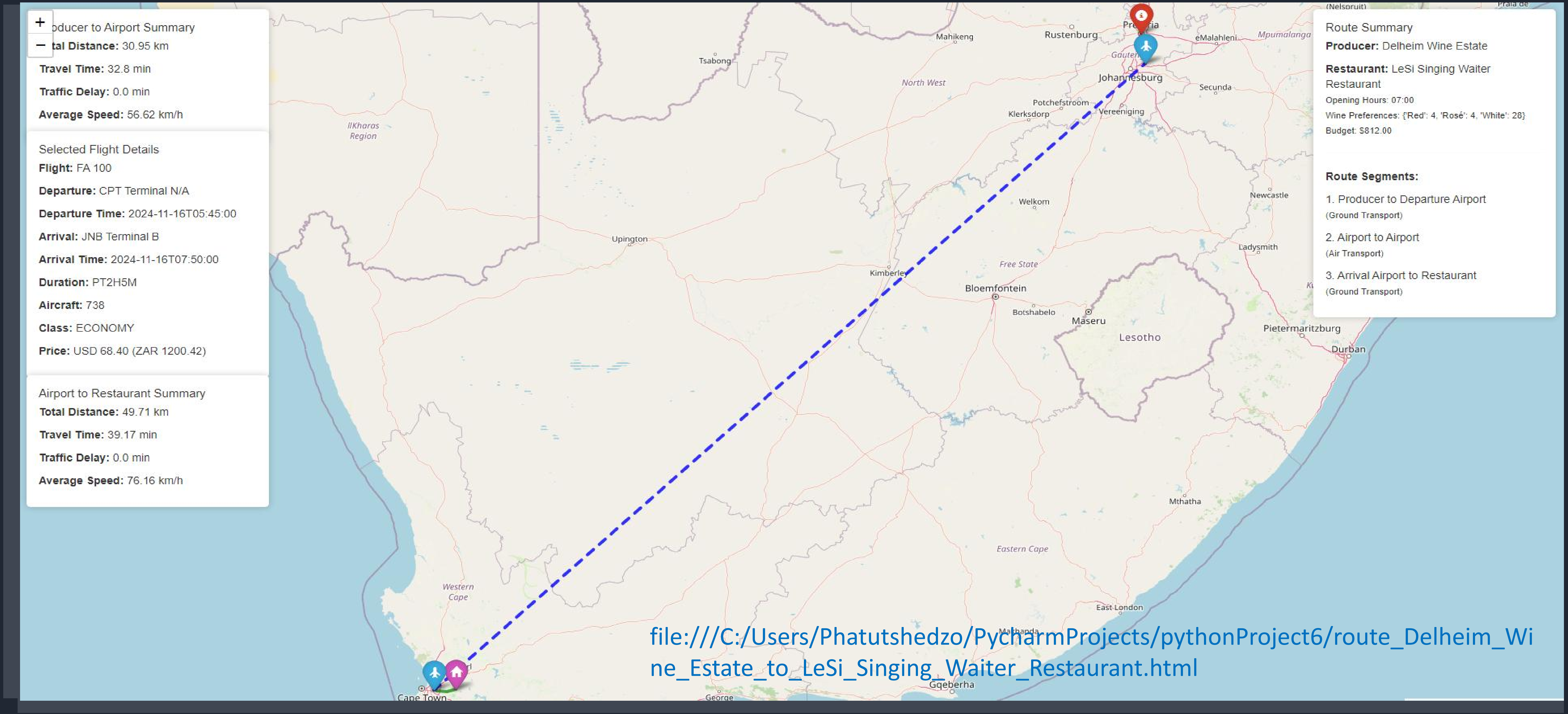


Demand Optimal Path & Financial Analysis



file:///C:/Users/Phatutshedzo/PycharmProjects/pythonProject6/route_Delheim_Wine_Estate_to_LeSi_Singing_Waiter_Restaurant.html

The Future of Wine Trading in South Africa

This platform empowers the South African wine industry, creating a more efficient and profitable wine trading ecosystem. By leveraging technology, the platform fosters innovation and growth, connecting producers with a global audience.